Santiago Páez Avendaño

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EDUCATION

PhD Student in Astrophysics — Mexico City, Mexico
Instituto de Astronomía, Universidad Nacional Autónoma de México (UNAM) — Aug. 2026 – Present

- Advisor: Dr. Yilen Gómez Maqueo Chew

M.Sc. in Astrophysics — Mexico City, Mexico
Instituto de Astronomía, Universidad Nacional Autónoma de México (UNAM) — Aug. 2024 – Aug. 2026

- Thesis: Data reduction method for OPTICAM multiband time series of transiting exoplanets
- Advisor: Dr. Yilen Gómez Maqueo Chew
- GPA: 9.44/10.0

B.A. in Natural Sciences Education — Chia, Colombia
Faculty of Education, Universidad de La Sabana — Jan. 2019 – Mar. 2024

- Awarded thesis: "Exoplanet Hackathon: A challenge-based learning proposal for the development of scientific skills".
- Advisor: Ph.D.(c) David Tovar Rodríguez
- GPA: 4.16/5.0

RESEARCH AND ACADEMIC EXPERIENCE

SCIENTIFIC ARTICLES

- **Páez**, **S.**, Gómez Maqueo Chew, Y., and Hebb, L. (2026). "Data reduction method for OPTICAM multiband time series of transiting exoplanets". *RAS Techniques and Instruments* 5, rzag021.
- Khandelwal, A. et al. [included **Páez, S.**] (2026). "TOI-7413 b: A super-Neptune in the Neptunian ridge orbiting and M dwarf". *in prep.*
- Khandelwal, A., Das, S. N., et al. [included **Páez, S.**] (2026). "TOI-6884b: a low-mass brown dwarf transiting a slightly evolved star". *Monthly Notices of the Royal Astronomical Society* 549(4), stag988.
- Gómez Maqueo Chew, Y. et al. [included **Páez, S.**] (2026). "TOI-1080 b: a temperate, rocky planet orbiting a quiet M4V host". *Monthly Notices of the Royal Astronomical Society* 548.
- **Páez**, **S.**, García, I., and Acevedo, A. (2022). "Enseñanza de la expresión genética: modelo y modelización en grado décimo". *Bio-grafía* 15(29), pp. 54–66.
- Leal Leal, M. et al. [included **Páez, S.**] (2021). "Mars Dichotomy: Prospects for human life on Mars". *EarthArXiv eprints* X51S5J.

ACCEPTED OBSERVING PROPOSALS AND OBSERVATIONAL EXPERIENCE

- **Observer**. SAINT-EX project. Time: 19 nights between Aug. 2026 and Oct. 2026 Instrument: SAINT-EX/OAN-SPM 1 m Telescope. Role: Launching remote observations.
- **Co-I**. Insights into stellar activity cycles: Spot crossing during transit. Time: 3 nights in Mar. 2025 and 3 nights in Mar. 2026 Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Data acquisition and data reduction.
- **Co-I**. Mass Measurements of Two Gas Giant Planets around M-dwarf stars. Time: 1.75 nights in 2026A. Instrument: WIYN/NEID. Role: Preparation of observing proposal and transit fitting.
- **Co-I**. Multi-band Transit Follow-up of Giant Planet Candidates Around M-Dwarfs with OPTICAM. Time: 6 nights in Feb. 2025 and 6 nights in Feb. 2026 Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Preparation of observing proposal, design of observation plan, data acquisition, data reduction, and transit fitting.

CONFERENCES, MEETINGS AND POSTERS

- **Páez**, **S.**, Pichardo Marcano, M., et al. Ground-based multiband transit follow-up with OPTICAM and SAINT-EX at OAN-SPM. *Poster at 2026 Sagan Exoplanet Summer Workshop*. Remote - Caltech, Pasadena, USA. 2026.
- Khandelwal, A., **Páez**, **S.**, et al. Validation of TOI-7413 b: A Neptunian Planet on the Neptunian Ridge Around an M Dwarf. *Poster at 2026 Sagan Exoplanet Summer Workshop*. Remote - Caltech, Pasadena, USA. 2026.
- **Páez**, **S.**, Gómez Maqueo Chew, Y., and Hebb, L. Data reduction method for OPTICAM multiband time-series of transiting exoplanets. *Oral presentation in the XXII Astronomy Students Meeting (REA)*. Mexico City, Mexico. 2025.
- **Páez**, **S.**, Gómez Maqueo Chew, Y., and Hebb, L. Método de reducción de datos para series de tiempo multibanda con OPTICAM de exoplanetas transitantes. *Seminar of Astrophysics and Cosmology COSMOS - Instituto de Ciencias Físicas - UNAM*. Online. 2025.
- **Páez**, **S.**, Tovar, D., and Barreto, C. Proyéctate: a novel strategy to reach the public audience using Astronomy as a catching science topic. Oral Presentation. *Communicating Astronomy with the Public - CAP*. 2024.
- León-García, A. et al. [included **Páez, S.**] Follow-up of WASP-43b and WASP-107b transits through differential photometry from OAN-TNT. Poster. *LXVI National Physics Congress*. Morelia, Mexico. 2023.
- **Páez**, **S.**, Barreto, C., and Rojas, N. Astronomy Festivals: A Strategy for Skill Development in Science Teacher Trainees. Oral Presentation. *XI National Congress on Physics and Astronomy Teaching*. Medellin, Colombia. 2022.
- **Páez**, **S.**, Puentes, G. A., and Becerra, D. Arduino Uno Models for Teaching about Exoplanets in Transit and Radial Velocity. Oral Presentation. *XI National Congress on Physics and Astronomy Teaching*. Medellin, Colombia. 2022.
- Tovar, D. et al. [included **Páez, S.**] Astrobiology Virtual Festival - Cota 2020. Novel strategies for astronomy, biology, and geology outreach in Cundinamarca (Colombia). Oral Presentation. *Communicating Astronomy with the Public - CAP 2021.* 2021.
- Tovar, D. et al. [included **Páez, S.**] Astronomy Festivals in Cundinamarca (Colombia) as Educational Strategies for Learning Scientific Concepts. Poster. *Communicating Astronomy with the Public - CAP.* 2021.

INTERNSHIPS, WORKSHOPS, AND SUMMER SCHOOLS

Code/Astro Workshop — Evanston, IL, USA

A week-long intensive training in astronomy software development. It covered the fundamental software engineering skills and best practices necessary for building sustainable open-source packages for astronomy applications, such as documentation, code profiling, unit tests, parallelization, code packaging and release. — Aug. 4–8, 2025.

District Education Secretariat — Bogota, Colombia

Intern with the Education Evaluation Department, International Baccalaureate Team. Provided support to public schools participating in the International Baccalaureate program. Assisted with logistics, operations, and digital tools for managing and organizing information. — Feb. 2024 – Jun. 2024.

XV ESAOBELA — Puebla, Mexico

100 hours of astronomical observation using 1-meter and 10-inch telescopes, as well as 51 hours of theoretical instruction at the Observatorio Astronómico Nacional de Tonantzintla (Puebla, México). — Jan. 2023 – Feb. 2023.

Abraham Maslow School — Chia, Colombia

Teacher Assistant for astronomy, biology, and physics classes, as well as a telescope operator. — Aug. 2023 – Nov. 2023.

Bogota's Planetarium — Bogota, Colombia

Evaluator of teaching strategies — Aug. 2022 – Nov. 2022.

XXXI Scientific Summer at the OAN-SPM Online
Summer school featuring theoretical classes on general astrophysics topics and remote observing sessions using the 84-cm telescope at the Observatorio Astronómico Nacional en la Sierra de San Pedro Mártir (Baja California, México). Jun. 2022.

Alpha-Cen Summer School Online
Summer School focused on developing programming skills for handling astronomical data and introducing core topics in astrophysics. Aug. 2021.

Bosques de Sherwood School Chia, Colombia
Teacher assistant in chemistry classes. Feb. 2022 – May 2022.

Secretariat of Science, Technology, and Innovation: Sopo Mayor's Office Sopo, Colombia
Astronautic festival organizer Aug. 2021 – Nov. 2021.

Interactive Museum Maloka Bogota, Colombia
Museum's guide and evaluator of teaching strategies. Feb. 2021 – May 2021.

RECOGNITIONS AND SCHOLARSHIPS

National Graduate Scholarship (Master's Degree) Mexico City, Mexico
Recipient of fully the funded National Graduate Scholarship by the Secretariat of Science, Humanities, Technology and Innovation (SECIHTI) of Mexico Aug. 2024 – Jun. 2026

Bogota City Council Bogota, Colombia
Recognition through an official council note for being a Top 50 Finalist of the Global Student Prize 2023 (Proposition No. 521 of 2023). Dec. 2023

Thesis IN-BITE Bogota, Colombia
2nd. Place in a thesis inter-university competition of the research group in physics of the Pedagogical National University (GRIFO) Nov. 2023

Best Saber Pro test score Chia, Colombia
Awarded for achieving the highest Saber Pro undergraduate examination results in the Natural Sciences Education program during 2022 Oct. 2023

Global Student Prize 2023 Varkey and Chegg.org Foundations
Top 50 Finalist. In recognition of exceptional achievements in the classroom and beyond. Only Colombian in the top 50 finalists out of 5,600 applicants worldwide. Jul. 2023

FEST Scholarship Bogota, Colombia
Fully funded undergraduate scholarship by Secretariat of Education of Bogotá Jan. 2019 – Nov. 2024

OUTREACH

Public Talk Mexico City, Mexico
"Ultra-precise light curves for studying transiting exoplanets. Clavius Astronomical Center of the Universidad Iberoamericana". Feb. 2026

Staff Noche de Las Estrellas Festival Mexico City, Mexico
Outreach staff in the Guillermo Haro Planetarium during the Noche de Las Estrellas Festival. Nov. 2025

Testimonial Journal UNAM International
"In search of exoplanets. The transit of a Colombian astronomer in Mexico". A testimonial describing the experience of pursuing a Master's degree in astrophysics at UNAM. Mar. 2025

Public Talk Orizaba, Mexico
"Mexico in the discovery and study of distant worlds". Orizaba Planetarium. Feb. 2025

Astronomy Outreach Festivals Sabana Centro Province, Colombia
Organized and participated in eight astronomy outreach festivals across multiple municipalities in the Sabana Centro province. Activities included workshops, public lectures, and nighttime observations, reaching over 4,000 participants in total. May 2019 – Nov. 2023

OTHER EXPERIENCES

Undergrads Tutor
Mentoring Program for new undergrad scholars at Universidad de La Sabana Aug. 2019 – Nov. 2023

PAT Program Member
Learning to Work Program at Universidad de La Sabana Aug. 2019 – Nov. 2023

- Support admissions program to the Faculty of Education
- Management of Proyéctate program

AstroSabana Group Leader
Study, outreach, and research students' group. Jan. 2019 – Nov. 2023

- Organization of outreach festivals.
- Management of astronomical observations for the public.
- Outreach astronomy talks to schools, shopping centers, and astronomy associations.
- Coordination of participating students.

LANGUAGES

- Spanish – Native
- English B2

SKILLS

Programming Python Git/GitHub MATLAB HTML/CSS R Studio

Software Jupyter Notebook AstroImageJ Poetry UV LaTeX Topcat VS Code

Scientific Packages `Astropy` `photutils` `Scipy` `juliet` `exoplanet` `pyaneti` `PyMC`

Operating Systems macOS Linux